

JOY SENGUPTA

MERN Stack Developer | Next.js Developer | Python Developer

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PROFESSIONAL SUMMARY

Motivated [B.Tech](#) AI & ML student at Narula Institute of Technology with hands-on experience in building **fullstack web applications** and **AI-driven tools**. Proficient in the **MERN stack** (MongoDB, Express.js, React.js, Node.js) and core ML frameworks including TensorFlow, PyTorch, and Scikit-learn. Proven track record of delivering **end-to-end projects** — from RESTful API integrations to intelligent automation systems — in **fast-paced internship environments**. Eager to contribute to product teams where strong engineering fundamentals meet cutting-edge AI.

WORK EXPERIENCE

Student Intern | Internship

Jan 2026 - Mar 2026

Eduskills

Remote • Kolkata

- Acquired skills in **Google AI/ML**.
- Learned **8-library ML stack through Google AIML training** — TensorFlow, Keras, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn — developing foundational models and data pipelines that cut data preprocessing time by 30% on academic datasets.

Technologies / Skills Used : TensorFlow, PyTorch, Scikit-Learn, Python

Gen AI Intern | Full-time

Jun 2025 - Sep 2025

AI Wallah

Remote • Kolkata

- Developed AI-powered full-stack applications with FastAPI + React, integrating external LLM/ML APIs to automate workflows and reduce manual effort by ~50%.
- Built and executed a 20+ case Postman test suite across REST endpoints — resolved 10+ pre-deployment bugs, delivering stable, regression-free releases.

Technologies / Skills Used : HTML, CSS, JavaScript, Python, FastAPI

PROJECTS

Purplexity | GitHub Website

Jun 2026 - Present

Lead Developer

- Built a **Perplexity-style AI answer engine** with a Next.js frontend and Bun/Hono backend, utilizing Openrouter for LLM inference.
- Implemented real-time SSE token streaming with **Zod-validated structured outputs** for reliable, progressive response rendering.
- Developed per-account daily token quota middleware and a multi-section Profile Settings module with session management, resolving a session-scoping bug in Active Sessions.
- Containerized the app as a multi-service Docker deployment behind an nginx proxy optimized for long-lived SSE connections.

Technologies / Tools Used : React, Node.js, Express, MongoDB, Tailwind CSS, Bun

VibeForge | GitHub Website

Jun 2026 - Jun 2026

Lead Developer

- Built an **AI-powered code review platform** that scans GitHub repos and pasted code for security issues, AI-generated "slop", code smells, and performance problems, scoring each codebase across 5 axes with an **A-F grade**.
- Developed a **Monaco-based web editor** with inline review decorations and AI fix suggestions, plus an npx CLI (`vibeforge scan`) featuring browser-based OAuth login and HTML/JSON report export.
- Implemented **Supabase auth** (Google + GitHub OAuth) with Prisma/Postgres, and a Razorpay subscription flow with free and Pro tiers.
- Architected a **Next.js 14 frontend** with a separate Express backend handling scan logic, billing, and LLM analysis through OpenRouter; deployed the web app on Vercel and published the CLI to npm.

Technologies / Tools Used : React, Node.js, Express, MongoDB, Tailwind CSS, Openrouter

Verifi | Website GitHub

May 2026 - Jun 2026

Lead Developer

- Built Verifi — a **full-stack misinformation detection platform** on microservices, processing text, URLs, and images via a **4-service agentic AI pipeline**.
- Engineered **Scout → Reader → Analyst pipeline** — autonomously extracts claims, crawls live evidence, and returns a cited veracity verdict in under 10s.
- Built the **React + Node.js frontend and API layer** — unified input interface for **text, URLs, and images** with real-time verdict rendering and source links.
- Demoed the **full agentic pipeline** live at **Innovexa 2026 (SurTech hackathon)** — presented end-to-end fact-checking flow under competition conditions.

Technologies / Tools Used : React, Node.js, Express, MongoDB, Tailwind CSS, Python, Flask

SKILLS

Programming Languages :	C, C++, Java, Python, JavaScript, TypeScript
Frameworks & Libraries :	React, Next.js, Node.js, Express, Flask, Tailwind CSS
Tools & Platforms :	Git, GitHub, Linux, Postman
Databases :	MongoDB, Supabase, Firebase
Soft Skills :	Communication, Leadership, Teamwork, Decision Making
Languages :	Hindi, English, Bengali

EDUCATION

Narula Institute of Technology | Kolkata

Aug 2024 - Jul 2028

B.Tech CSE(AIML)

Pursuing a Bachelor of Technology in **Artificial Intelligence** and **Machine Learning**. Developing a strong foundation in **computer science** principles, advanced **mathematics**, and modern AI algorithms.